

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS:

Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

- A. `<media src="video.mp4"></media>`
B. `<video src="video.mp4" controls></video>`

- C. `<movie src="video.mp4"></movie>`
- D. `<vid src="video.mp4"></vid>`

Answer: B. `<video src="video.mp4" controls></video>`

Justification:

The `<video>` tag is the standard HTML5 element used to embed videos in a web page. The `controls` attribute provides built-in playback controls such as play, pause, volume, and full screen. The other tags (`<media>`, `<movie>`, and `<vid>`) are not valid HTML5 elements.

Snippet 2: Form Submission Methods

- A. GET
- B. POST
- C. PUT
- D. DELETE

Answer: B. POST

Justification:

The POST method sends form data in the request body, making it more secure than GET for submitting sensitive information such as usernames, passwords, and registration details. It is the standard method for registration forms.

Snippet 3: Navigation Elements

- A. `<div>`
- B. `<nav>`
- C. `<section>`
- D. `<header>`

Answer: B. `<nav>`

Justification:

The `<nav>` element is the semantic HTML tag used to define a section containing navigation links. It improves website accessibility, readability, and search engine optimization (SEO). The `<div>` element has no semantic meaning, while `<section>` and `<header>` are used for different purposes.

Snippet 4: XML Syntax

- A. `<note><to>User</to><from>Admin</from>`
- B. `<note><to>User</to><from>Admin</from></note>`
- C. `<note><to>User</from></note>`
- D. `<note to="User"></note>`

Answer: B. `<note><to>User</to><from>Admin</from></note>`

Justification:

This is the correct XML syntax because all tags are properly opened, nested, and closed.

Errors in the other options:

Option A: Missing the closing `</note>` tag.

Option C: Incorrect closing tag (`</from>` instead of `</to>`).

Option D: Missing the closing `>` in the `</note>` tag, making it invalid XML.

Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

Answer: B. HTTP

Justification:

HTTP (Hypertext Transfer Protocol) is the standard protocol used for communication between web browsers and web servers. FTP is used for file transfer, SMTP is used for email transmission, and SNMP is used for network management.

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

I Enamala Gowtham/192311190(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Enamala Gowtham

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand